for intangible, private events such as expectations and emotions (Levine, 1997). After people find out how an event turns out, they remember best the previous facts that fit with the final outcome. From the facts that stand out in their memory, they reconstruct what they "must have" previously thought.

SOMETHING TO THINK ABOUT

Can you interpret people's beliefs that they had a "psychic hunch" in terms of hindsight bias? ✸

Eyewitness Identifications

Suppose you witness a crime: several hours, days, or weeks later, the police ask you to identify the culprit. You still remember the face somewhat, but your memory has faded. To help you, the police offer you a lineup of five or so possible suspects and ask whether you recognize any of them as the guilty party. Might you make a mistake?

Psychologists have conducted studies with staged crimes and various kinds of lineups. They find that witnesses are not always accurate. Witnesses who express higher confidence tend to be more accurate, although they make mistakes too (Sporer, Penrod, Read, & Cutler, 1995). When the police try to develop an unbiased lineup, they often start with someone they suspect and then add others who look similar. (If the witness described a young man with a mustache, it would obviously be unfair to set up a lineup with one young, mustached man and four older, clean-shaven men.) Unfortunately, a lineup with the main suspect and others chosen because they resemble the suspect is also biased, for a subtler reason (Wogalter, Marwitz, & Leonard, 1992): If all the others are selected because they resemble the suspect, the suspect stands out as the one who is *most similar* to all the others in the group! For

TRY IT YOURSELF

instance, examine the lineup in Figure 7.26. Without even having witnessed the crime, would you be more likely to pick out one of these people than the others? (Check answer D, page 265.) It is difficult to make a lineup fair (Wells, 1993; Wogalter, Marwitz, & Leonard, 1992). The point is that, because memory is so suggestible, it is difficult to ask a totally unbiased, nonsuggestive question.

The "False Memory" Controversy

Occasionally, someone tells a therapist about vague unpleasant feelings, and the therapist replies, "Symptoms like yours are usually found among people who were abused, especially sexually abused, in childhood. Do you think you were?" In some cases, the client says yes; in other cases, the client says no, and the therapist accepts the denial and moves on to other topics. A major controversy has arisen, however, about cases in which a therapist does not accept "no" for an answer: "The fact that you can't remember doesn't mean that it didn't happen. It may have been so painful that you repressed it." The therapist may then recommend hypnosis, repeated attempts to remember, or other techniques. Several sessions later, the client may say, "It's starting to come back to me . . . I think I do remember . . . ." *Reports of long-lost memories, prompted by clinical techniques,* are known as **recovered memories.**

Sexual abuse in childhood does occur, and no one knows how often. Some abused children do develop long-lasting psychological scars. But when people claim to recover long-forgotten memories, should we assume their reports to be accurate? Some reports are bizarre. In one case, two sisters accused their father of raping them many times, both vaginally and anally, bringing his friends over on Friday nights to rape them, forcing them to participate in satanic rituals that included cannibalism, the slaughter of babies, and so forth (Wright, 1994). The sisters had not remembered any of these events until they spoke with a therapist. In another case, a group of 3- and 4-year-old children, after urgings from a therapist, accused their Sunday school teacher of sexually abusing them with a curling iron, forcing them to drink blood and urine, hanging them upside down from a chandelier, dunking them in toilets, and killing an

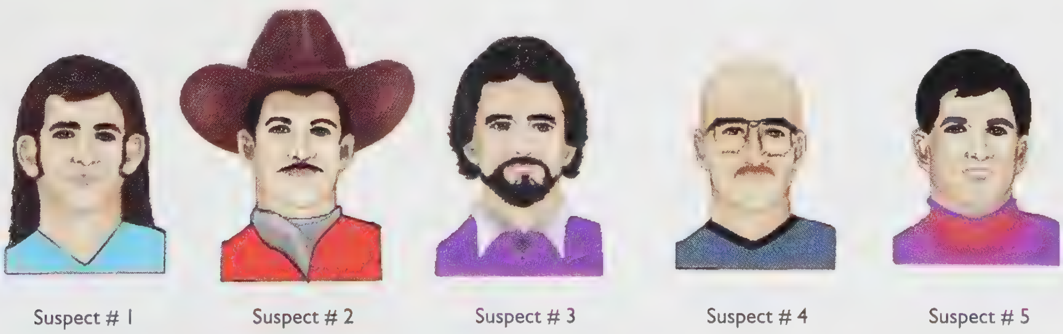


FIGURE 7.26 Here, a lineup is composed of one "suspect" and others chosen to resemble the suspect. Does one of these stand out as the probable suspect? (Check answer D, page 265.)

elephant and a giraffe during Sunday school class (Gardner, 1994). In neither of these cases, nor in several similar cases, was there any physical evidence to support the claims—scarred tissues or giraffe bones, for example.

Recovered-memory claims are not always so bizarre, but their accuracy is almost always uncertain. Suppose a 30-year-old woman claims that her father sexually abused her when she was 10. She had not remembered it until now, when her therapist repeatedly asked her to imagine "what it would have been like if she had been abused." In almost all cases, it is impossible to check the accuracy of such recovered memories. One might consult school records of long ago to see whether they described a happy, healthy child or one showing physical and emotional scars, but even the best of such records are incomplete and possibly misleading. Psychological researchers can, however, address these questions: When people have miserable, abusive experiences, are they likely to forget them for a long time? And is it possible to persuade someone to "remember" an event that never actually happened, just by suggesting it?

Repression

Sigmund Freud, whom we shall consider more fully in Chapter 13, introduced the term **repression** as *the process of moving a memory, motivation, or emotion from the conscious mind to the unconscious mind.* Although many therapists continue to use the concept of repression, the research on memory and forgetting has not supported it (Holmes, 1990). Controlled experiments designed to produce repression have produced small, inconsistent, ambiguous effects.

Other studies have examined people known to have had dreadful experiences. One study examined 16 children who had witnessed the murder of one of their parents. All had recurring nightmares, haunting thoughts, and painful flashbacks of the experience; none showed any indication of forgetting or repressing the memories (Malmquist, 1986). Another study examined 20 children who, at ages less than 5 years old, had survived a plane crash, or had been kidnapped, or had been forced to participate in the filming of pornographic movies. Of those who were at least 3 years old at the time of the trauma, 6 out of 9 had good recall of the events even years later, and the other 3 had partial memories—in contrast to the poor recall of most early childhood experiences (Terr, 1988). From such results, we cannot conclude that repression never occurs. (It is almost impossible to prove that something *doesn't* exist. Could you prove that unicorns don't exist?) However, the harder it is to demonstrate a phenomenon, the rarer it must be in real life—if it exists at all.

People do *forget* some painful memories, especially from early childhood, if given enough time. But because we are as likely to forget happy memories as unhappy ones—maybe even more likely—we should not attribute our memory loss to repression.

WHAT'S THE EVIDENCE?

Suggestions and False Memories

Critics of the concept of recovered memories have suggested that a therapist, by repeatedly suggesting that a client recall certain kinds of memories, can unintentionally "implant" a **false memory,** *a report that someone believes to be a memory but that does not actually correspond to real events* (Lindsay & Read, 1994; Loftus, 1993). Researchers cannot determine whether a particular report of recovered memory was true or false, but they can test how easy it is to implant a false memory. We shall examine three representative experiments.

EXPERIMENT 1

Hypothesis If people are asked questions that suggest or presuppose a certain fact, many people will later report remembering that "fact," even if it never happened.

Method Elizabeth Loftus (1975) asked two groups of students to watch a videotape of an automobile accident. Then she asked one group but not the other, "Did you see the children getting on the school bus?" In fact, the videotape did not show a school bus. A week later, she asked both groups 20 new questions about the accident, including this one: "Did you see a school bus in the film?"

Results Of the first group (those who were asked about seeing children get on a school bus), 26% reported that they had seen a school bus; of the second group, only 6% said they had seen a school bus.

Interpretation The question "Did you see the children getting on the school bus?" presupposes that there *was* a school bus in the film. Some of the students who heard that question reconstructed what happened by combining what they actually saw with what they believed might reasonably have happened and what someone (the researcher) had suggested to them afterward.

EXPERIMENT 2

The first experiment demonstrated that a suggestion can distort memories of something that people watched. But could a suggestion also distort a personal memory, or even implant a memory of an experience that never really occurred?

Hypothesis If people are told about a childhood event by people they trust, they will come to remember it as something they actually experienced, even if in fact they did not.

Method The participants, aged 18 to 53, were told that the study concerned their memories of childhood. Each was given four paragraphs; each paragraph described a different event. Three of the events had actually happened.

(The experimenters had contacted parents to get descriptions of childhood events.) A fourth event was a plausible story about getting lost, which had not actually happened, described in terms of family members. An example for one Vietnamese woman: "You, your Mom, Tien and Tuan, all went to the Bremerton K-Mart. You must have been five years old at the time. Your Mom gave each of you some money to get a blueberry ICEE. You ran ahead to get into the line first, and somehow lost your way in the store. Tien found you crying to an elderly Chinese woman. You three then went together to get an ICEE."

After reading the four paragraphs, each participant was asked to write whatever additional details they could remember of the event. They were asked to try again one week later, and then again after another week.

Results Of 24 participants, 6 reported remembering the suggested (false) event. Participants generally described the false memories in fewer words than the correct memories, but some did provide a fair amount of "additional detail." The woman in the example above said, "I vaguely remember walking around K-Mart crying and looking for Tien and Tuan. I thought I was lost forever. I went to the shoe department, because we always spent a lot of time there. I went to the handkerchief place because we were there last. I circled all over the store it seemed 10 times. I just remember walking around crying. I do not remember the Chinese woman, or the ICEE (but it would be raspberry ICEE if I was getting an ICEE) part. I don't even remember being found" (Loftus, Feldman, & Dashiell, 1995).

Interpretation A simple suggestion can provoke some people to recall a personal experience in moderate detail, even though the event never happened. Granted, the suggestion affected fewer than half of the people tested, and most of them reported only vague memories. Still, the researchers used only a brief suggestion, without hypnosis or other techniques that probably would have increased the effect.

EXPERIMENT 3

Defenders of the concept of recovered memories are not impressed with false memories such as being lost at a Kmart. Even if this woman was not lost at Kmart at age 5, she was probably lost somewhere else at some other time. Implanting a false memory of such a likely event does not demonstrate that one could implant a false memory of, say, childhood sexual abuse.

To this criticism, researchers reply that it would be unethical to try to implant memories of something so traumatic. However, it is possible to compare the difficulty of implanting a memory of a reasonably likely event with an implanted memory of an implausible one.

Hypothesis When adults hear a suggestion about a childhood religious experience, they will be more likely to accept it and claim to remember it if it pertains to the kind of ceremony they practice in their own religion. They will probably not accept a suggestion about a ceremony from another religion.

Method The experimenters began by contacting the mothers of 29 Roman Catholic and 22 Jewish teenage girls. All had been active in their religious practices through childhood and into the present. The mothers provided descriptions of events (not necessarily religious) that actually happened to their daughters at about age 8.

The experimenters told each girl five brief episodes and asked them to provide any additional details they could remember. Three of these were real events, as described by the mothers. Another one was a false event that would be plausible to the Catholic girls but not to the Jewish girls: taking Communion and returning to the wrong seat. The other was a false event that should be plausible to the Jews but not to the Catholics: conducting Friday night prayers before sunset and dropping the bread.

Results Most girls remembered all three of the correct events and provided additional information. Here are the results for the false events (Pezdek, Finger, & Hodge, 1997):

	Recalled neither	Recalled the Catholic event	Recalled the Jewish event	Recalled both
Catholic girls (total of 29)	19	7	1	2
Jewish girls (total of 22)	19	0	3	0

Interpretation As predicted, it was easier to implant plausible than implausible memories, but three of the Catholic girls did report false memories corresponding to a Jewish ceremony. Evidently, the more likely a suggested event and the more similar it is to one's actual experience, the easier it is to implant a false memory. But there is probably no sharp line between implantable and nonimplantable memories.

So, what conclusion should we draw about the possibility of implanting false memories of sexual abuse and the like? The experiments described here are just examples of a large body of research, but they all follow about the same pattern: Researchers use either a single suggestion or a few repetitions. The more repetitions, the bigger the effect (Zaragoza & Mitchell, 1996). The suggestions pertain to nontraumatic events, and the participants are normal people with no known psychological problems. In contrast, therapists who are seeking recovered memories use repeated suggestions, sometimes accompanied by hypnosis or other techniques. They suggest highly traumatic experiences, and they deal with troubled people. Obviously, we cannot directly extrapolate from one to the other. In particular, the research evidence does not demonstrate that all recovered memories must be false (Pope, 1996).

On the other hand, it is clearly unfair to dismiss the research simply because it has not demonstrated false memories of sexual abuse or satanic rituals. Ethically, researchers cannot attempt such, and they should not need to do so. If you can build a small fire with a few twigs, you shouldn't have to demonstrate that, with more fuel, you could build a bigger fire.

The main conclusion is that a therapist, police investigator, or anyone else who repeatedly suggests that something "might have" or "must have" occurred runs a major risk of implanting or distorting a memory. People who are seeking the truth need to become more aware of this risk and more cautious about the questions they ask. Furthermore, if someone claims to have recovered a long-forgotten memory, other people should treat it as no more than a hypothesis, one that might be true but could be false. Such testimony, by itself, in the absence of supporting evidence, should not be a basis for making an arrest, breaking up a family, or taking any other drastic action.

* * *

Children as Eyewitnesses

Finally, what about children who are witnesses or victims of a crime? Young children forget more rapidly than adults do, and they sometimes confuse fantasy with reality. Are they therefore unreliable as witnesses?

They do tend to be more vulnerable than adults are to suggestion, and they can quickly become unreliable witnesses if the interviewers ask the wrong kind of questions. In one study, preschool children were repeatedly asked to "think about" getting their hand caught in a mousetrap. Eventually, most of them provided elaborate details about the injury and the ensuing trip to the hospital. They provided as much detail about this false memory as about a real injury, and they gave every indication of believing their reports (Ceci, 1995).

However, in two other studies, children were asked about the events of their medical examinations at delays ranging from 3 days to 6 weeks after the event. Children of ages 3 to 7 reported the events accurately and in detail. Researchers then asked suggestive or leading questions such as "Did the nurse cut your hair?" and "Did the person hit you?" Even at younger ages, only a few children reported events that had not happened (Baker-Ward, Gordon, Ornstein, Larus, & Clubb, 1993; Goodman, Aman, & Hirschman, 1987). These results, of course, do not deny that one could induce children to give false testimony by more prolonged and insistent suggestions, but they do indicate that children can give accurate reports under favorable circumstances.

The general recommendations for children's eyewitness testimony are as follows: If a child is simply asked to describe the events, in a nonthreatening atmosphere, without suggestions or pressure, reasonably soon after the event, even children as young as 3 can be believable (Ceci & Bruck, 1993). Asking a child the same questions repeatedly within a single interview, though, can confuse the child. (Some will change their answers on the assumption that their first answer must have been wrong.) However, asking the child to repeat the story every few days may help the child to remember; this may be necessary if the child is to testify in a trial many months later (Poole & White, 1995).

SOMETHING TO THINK ABOUT

Some people claim to have been abducted by aliens from another planet. Presuming that they are not deliberately lying to get attention, that they are not mentally ill, and that they were not in fact abducted by aliens, how might we explain their reports? ✸

THE MESSAGE
Memories, True, False, and Uncertain

A historian who writes about times long ago reconstructs events based on the remaining records and the historian's inferences about what must have happened. We do the same with our own memories. If someone suggests that "surely you remember the time when you got lost at Kmart . . . ," you might reconstruct a memory of something that did not actually happen. Have you ever reminisced with someone about an event from years ago, only to discover that the two of you remembered that event very differently? If you want to record an event accurately forever, take notes or photographs. Your notes or photos will fade, too, but at least they won't be transformed into something entirely different.

He: *We met at nine.*
She: *We met at eight.*
He: *I was on time.*
She: *No, you were late.*
He: *Ah, yes! I remember it well.*
We dined with friends.
She: *We dined alone.*
He: *A tenor sang.*
She: *A baritone.*
He: *Ah, yes! I remember it well.*
That dazzling April moon!
She: *There was none that night.*
And the month was June.
He: *That's right! That's right!*
She: *It warms my heart to know that you*
Remember still the way you do.
He: *Ah, yes! I remember it well.*

—"I REMEMBER IT WELL" FROM THE MUSICAL *GIGI* BY ALAN JAY LERNER AND FREDERICK LOEWE

SUMMARY

* *Reconstruction.* When remembering stories or events from their own lives, people recall some of the facts and fill in the gaps, based on logical inferences of what must have happened. They rely particularly heavily on inferences when they are uncertain about their memory of the events. (page 258)
* *Reconstructions from a word list.* If people read or hear a list of related words and try to recall them, they are likely to include closely related words that were not on the list. (page 258)
* *Hindsight bias.* People often revise their memories of what they previously expected, saying that how events turned out was what they had expected all along. (page 259)
* *Eyewitness testimony.* Eyewitness testimony is sometimes accurate, sometimes not. Lineups can bias a witness toward identifying a particular suspect. (page 261)
* *False memory debate.* Some therapists have used hypnosis or very suggestive lines of questioning to try to help people remember painful experiences. Some of the reported memories include extreme or bizarre events that people claim they experienced long ago but did not remember until much later. It is important to try to distinguish actual cases of abuse from false memories that depend on suggestion. (page 261)

Suggestion for Further Reading

Schacter, D. (Ed.) (1995). *Memory distortion.* Cambridge, MA: Harvard University Press. Collection of chapters about false memories and other kinds of distorted memories.

Terms

reconstruction putting together an account of past events, based partly on memories and partly on expectations of what must have happened (page 258)

hindsight bias the tendency to mold our recollection of the past to fit how events later turned out (page 260)

recovered memory a report of a long-lost memory, prompted by clinical techniques (page 261)

repression according to Freudian theory, the process of moving a memory, motivation, or emotion from the conscious mind to the unconscious mind (page 262)

false memory a report that someone believes to be a memory but that does not actually correspond to real events (page 262)

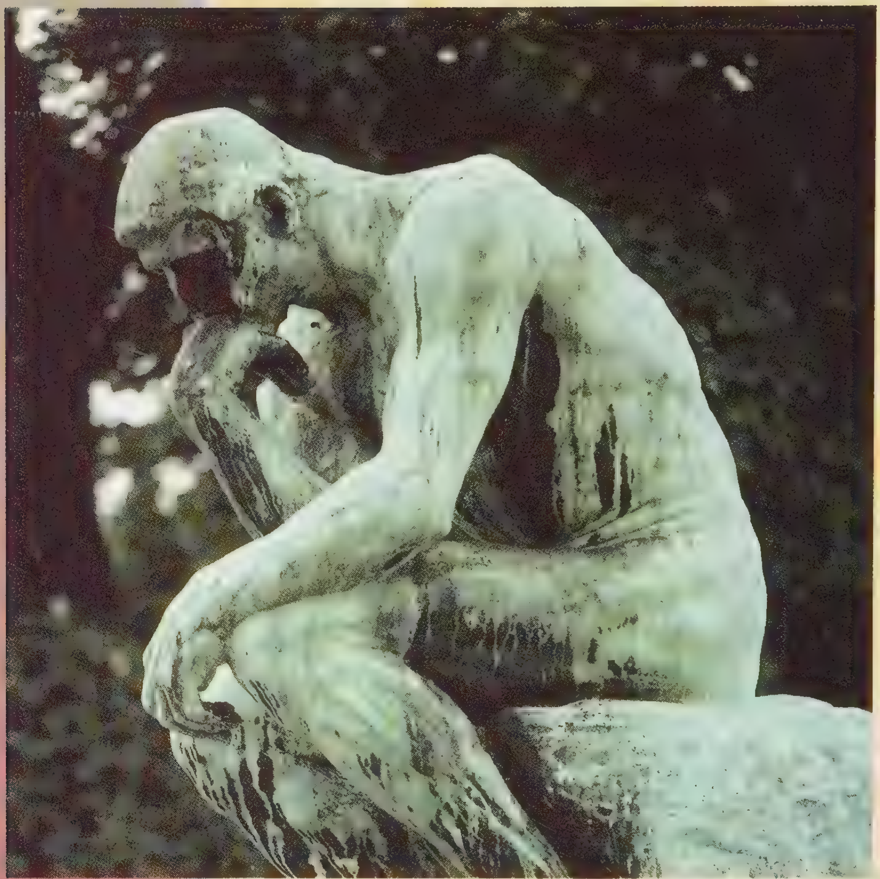
Answer to Question in the Text

D. Face #5 is the suspect. All the others have been chosen to resemble him, but each has a different unusual feature. (page 261)

Web Resources

False Memory Syndrome Foundation home page
advicom.net/~fitz/fmsf/
The scientific community in general does not support the notion that recovered memories are reliable. *The Skeptic* magazine addresses the issue; the APA delivers a policy statement and a Q&A about memories of childhood abuse.

Cognition and Language



8

MODULE 8.1
Thinking and Mental Processes

MODULE 8.2
Problem Solving, Expertise, and Error

MODULE 8.3
Language

Consider the statement, "This sentence is false." Is the statement itself true or false? Declaring the statement to be true agrees with its own assessment that it is false. But declaring it to be false would make its assessment correct.

A sentence about itself, called a *self-referential* sentence, can be utterly confusing, like the one above. It can also be true (e.g., "This sentence consists of six words"), false ("Anyone who reads this sentence will be suddenly transported to the planet Neptune"), untestable ("Whenever no one is reading this sentence, it changes into the passive voice"), or amusing ("This sentence no verb" or "This sentence sofa includes an unnecessary word").

In this chapter, you will be asked to think about thinking, talk about talking, and read about reading. Doing so is self-referential, and if you try to "think about what you are thinking now," you can easily go into a confusing loop similar to the one in "This sentence is false." For that reason, among many others, psychological researchers focus as much as possible on results obtained from carefully controlled experiments, not just on what people say they think about their own thought processes.

Cognition includes both old knowledge (such as knowing how to read a blueprint) and the use of that knowledge in a new situation.

MODULE 8.1
Thinking and Mental Processes

How do we think about categories of objects?

How accurate are our mental images and maps?

To what extent can we control our attention, and do certain stimuli capture it automatically?

Cognition is psychologists' word for ***thinking, gaining knowledge, and dealing with knowledge.*** Cognitive psychologists study how people think, how they acquire knowledge, what they know, how they imagine, and how they solve problems. They also deal with how people organize their thoughts into language and communicate with others.

Perhaps it seems to you that cognitive psychology should be trivially simple. "If you want to find out what people think or what they know, why not just ask them?" Sometimes psychologists can and do ask them; however, in many cases, people are not fully aware of their own thought processes (Kihlstrom, Barnhardt, & Tataryn, 1992). Recall, for example, implicit memory, as discussed in Chapter 7: Sometimes you see or hear something that influences your behavior in the next minute or so, without your realizing it. Similarly, we often solve a problem so fast that we do not know how we did it.

Cognitive psychology increased in popularity during the 1950s through the 1970s, the era of vastly increased use of computers. Although brains and computers do not work the same way, computers provide a valuable way of modeling theories of cognitive processes. A researcher may say, "Imagine that cognitive processes work as follows. . . . Now let's program a computer to go through those same steps in the same order. If we then give the computer the same information that a human has, will it draw the same conclusions a human does and make the same errors?" In short, computer modeling provides a good method to test theories of cognition. Today, cognitive psychology uses a variety of methods to measure mental processes and to test theories about what we know and how we know it.

SOMETHING TO THINK ABOUT

The Turing Test, suggested by computer pioneer Alan Turing, proposes the following operational definition of artificial intelligence: A person poses questions to a human source and to a computer, both in another room. The human and the computer send back typewritten replies, which are identified only as coming from "source A" or "source B." If the questioner cannot determine which replies are coming from the computer, then the computer has passed a significant test of understanding.

Suppose a computer did pass the Turing Test. Would we then say that the computer "understands," just as a human does? Or would we say that it is merely mimicking human understanding? ✷

Categorization

An ancient Greek philosopher once wrote that we can never step into the same river twice. In a sense, of course, he was right; everything is constantly changing. However, we generally find it useful to think of the Nile as "the same" river from one minute to the next, even from one century to the next. It also suits our purposes to use "river" as a general concept, lumping together the Nile, the Amazon, the Mississippi, and thousands of others, despite their differences.

Forming useful categories enables us to make educated guesses about features we have never seen for ourselves (Anderson, 1991). For example, your concept of a river enables you to infer that a river that you have never seen probably has many of the features of familiar rivers, such as a flow of water, navigability by boats, a certain degree of curviness, and the presence of fish.

We often take our categories for granted, as if our own way of categorizing objects were the only possible way (Figure 8.1). But people in other cultures sometimes use categories that seem strange to us (Lakoff, 1987). The Japanese word *hon* refers to sticks, pencils, trees, hair, hits in baseball, shots in basketball, telephone calls, television programs, a mental contest between a Zen master and a student, and medical injections. Clearly, people from different cultures categorize objects in different ways.

How do people decide how to categorize objects? To understand thinking, we need to understand how we categorize.

Categorization by Prototypes

If you look up a word such as *river* in a dictionary, you will find a definition. Do we also look up our concepts in a mental dictionary to determine their meaning? In some cases, probably yes. For example, we think of the term *bachelor* as an unmarried male. Because we would not ordinarily apply the term bachelor to a 3-year-old boy or to a Catholic priest, we might want to modify the definition to "a male who has never married but who could decide to get married." Such a definition pretty well explains the concept of a bachelor.

Few concepts can be so clearly defined, however. More frequently, we deal with loosely defined categories such as

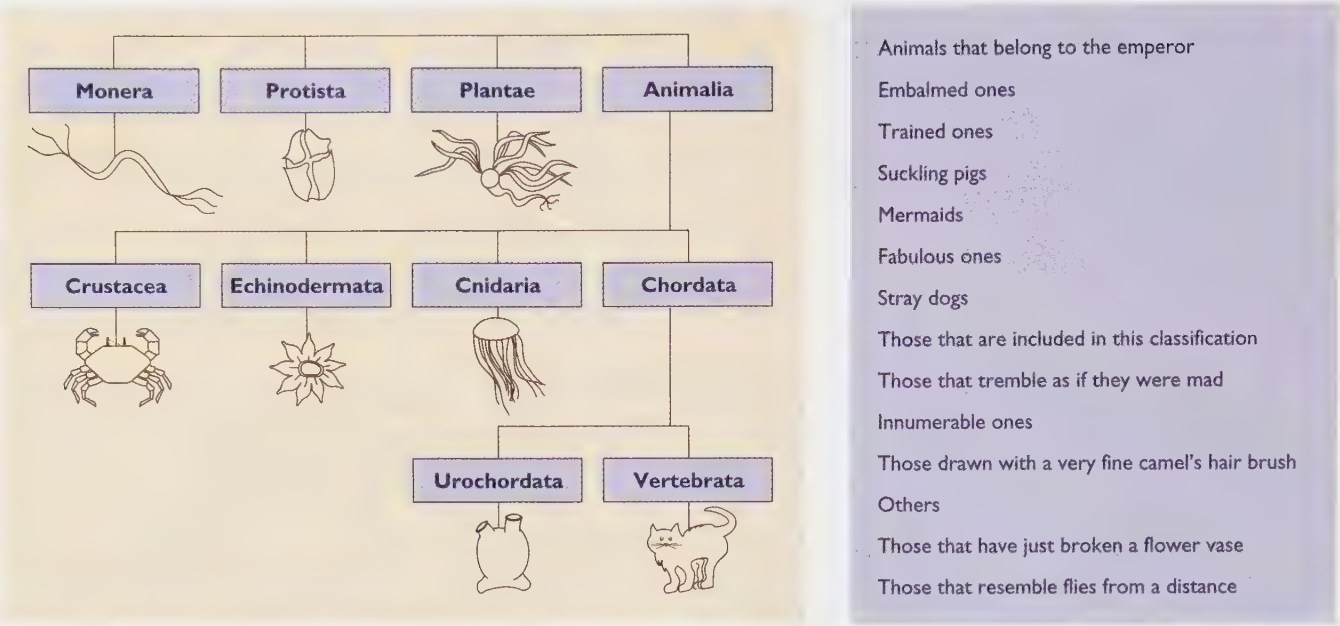


FIGURE 8.1 Left: A much-abridged chart of the current scientific classification of the animal kingdom. Right: An alleged listing from an ancient Chinese encyclopedia—actually the product of someone's imagination (Rosch, 1978). The point is that there are many ways to categorize animals or anything else, and some methods of categorizing are better than others.

interesting novels or embarrassing experiences, in which we could not list the defining features. Even for such a simple, everyday concept as a table, we have trouble describing the necessary and sufficient features: Most tables are made of wood, but they need not be. Most have four legs, but they can have more or fewer. The top surface must be reasonably flat, but not everything with a flat surface is a table.

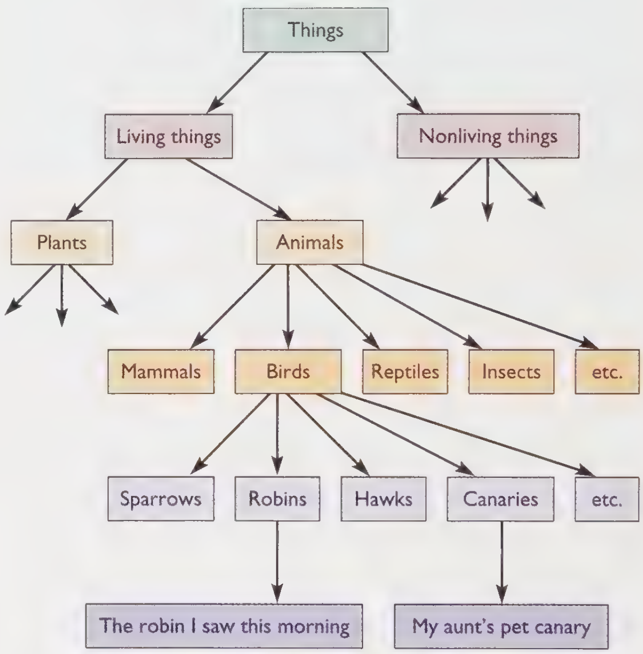
According to Eleanor Rosch (1978; Rosch & Mervis, 1975), many categories are defined by *familiar or typical examples* called **prototypes.** According to the categorization by prototypes approach, we decide whether an object belongs to a category by determining how well it resembles the protypical members of that category.

For example, we define the category *vehicle* by giving examples: *car, bus, train, airplane, boat.* To decide whether another object is a vehicle, we compare it to these examples. If we ask people whether a truck is a vehicle, they quickly respond "yes." They have longer reaction times to the atypical example *blimp* and still longer reaction times to *elevator* or *water skis.* The main point of Rosch's prototype approach is that category membership can be a matter of degree.

Conceptual Networks

TRY IT YOURSELF Choose any word and try to think about just that one word, nothing else, for 30 seconds. Why is this task so difficult? Because any word reminds you of something else. You do not store words in your brain in isolation, but in networks of related ideas.

First, you might link your word to subcategories and supercategories. For example, you link the word *bird* to more general and more specific terms as follows:



Researchers can demonstrate the reality of this kind of hierarchy (Collins & Quillian, 1969, 1970). Suppose you are asked several true-false questions about canaries:

Canaries are yellow.
Canaries sing.
Canaries lay eggs.
Canaries have feathers.
Canaries have skin.

Almost everyone will correctly answer "true" to all five items. As in many cognitive psychology experiments, the dependent variable is the reaction time (speed of answering). If you are like most people, your response is fastest on the *yellow* and *sing* items, a bit slower (in milliseconds) on the *eggs* and *feathers* items, and still slower on the *skin* item. Why? It is because yellowness and singing are distinctive, well-known characteristics of canaries. You probably do not think of eggs or feathers as specific features of canaries; instead you reason (quickly), "Canaries are birds, and birds lay eggs. So canaries lay eggs." Skin is not a particularly distinctive feature of canaries or even of birds, so you have to reason, "Canaries are birds and birds are animals. Animals have skin, so canaries must have skin."

Even though this way of categorizing things causes you slight delays in answering certain questions, such as whether canaries have skin, it actually saves you time and effort in the long run. When you learn that all birds have body temperatures above 40° C., you don't have to learn this fact again separately for every species of bird. When you learn that all animals have mitochondria in their cells, you don't have to relearn that fact for every animal species. Thus, reasoning in terms of categories and subcategories makes our lives easier.

CONCEPT CHECK

1. Which would take longer to answer: whether fashion models wear dresses, or whether fashion models sometimes get sick? Why? (Check your answer on page 278.)

We also link a word or concept to other concepts that relate to it in a variety of ways. Figure 8.2 shows a possible network of conceptual links for one person (Collins & Loftus, 1975). The links will, of course, vary from one person to another and from one moment to another. Suppose this network describes your own concepts. ***When you hear about or think about one of the concepts shown in this figure, exciting that concept will activate, or prime, the concepts linked to it*** (Collins & Loftus, 1975). The process of doing so is called **spreading activation.** As a result of spreading activation, thinking of one concept makes it easier to think of its related concepts also. For example, if you hear the word *flowers,* you are temporarily more likely than usual to think of the word *roses.* If you hear *flowers* and then see the word *roses* flashed briefly on a screen or hear it spoken very softly, your probability of identifying *roses* correctly will be higher than usual. Spreading activation can also combine from two sources. For

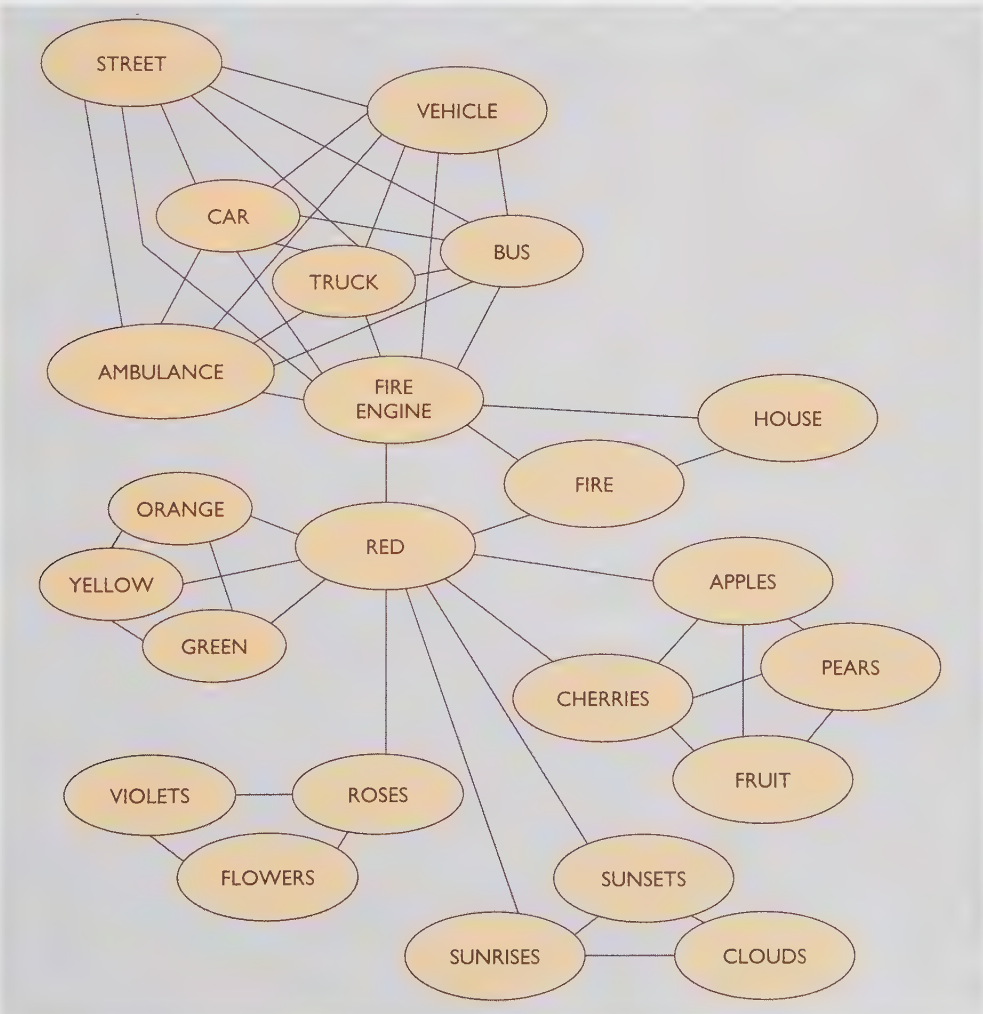


FIGURE 8.2 We link each concept to a variety of other related concepts. Any stimulus that activates one of these concepts will also partly activate (or "prime") the ones that are linked to it. (From Collins & Loftus, 1975.)

example, if you hear the words *flowers* and *red*, you have an even higher probability of thinking of the word *roses*.

As this theory would predict, a concept that is linked to many other concepts will activate each of them only slightly, whereas a concept that is linked to only a few will activate them more strongly. For example, *fruit* is linked with a relatively small number of items—more than shown in Figure 8.2, but still not an enormous number. The concept *red* is linked to a much larger number of other items. Suppose we ask various people the following questions and then measure their speed of response:

> What is something red that is a fruit?
> What is something that is a fruit and is red?

On average, people respond slightly faster to the second question, and the spreading-activation theory explains why (Collins & Loftus, 1975). With the first question, red produces only a small spreading activation for *apple* (the most likely correct answer). So the person answering the question does not get much of a start on this question until hearing the word *fruit*. With the second question, the word *fruit* provides a good start and has significantly activated *apple* before the person hears *red*.

CONCEPT CHECK

2. Would people respond faster to naming "something that makes sounds and tells time" or "something that tells time and makes sounds"? (Check your answer on page 278.)

Mental Imagery

When people think about three-dimensional objects or about places where they have been, do their mental images actually resemble vision? You might think this is a trivial question that does not need to be researched: "Of course mental images are like real vision," you might reply. "I see mental images all the time." Your self-reports are not solid evidence, however. After all, people sometimes insist that they have a clear mental image of an object but then find that they cannot correctly answer simple questions about it.

To illustrate: Imagine a simple cube balanced with one point (corner) on the table and the opposite point straight up. Imagine that you are holding the highest point with one finger. Now, with a finger of the opposite hand, point to all the remaining corners of the cube. How many corners do you touch?

TRY IT YOURSELF

You probably will say that you answered this question by "picturing" a cube in your mind as if you were actually seeing it. However, most people answer the question incorrectly, and only a few get the right answer quickly (Hinton, 1979). (Check answer A, page 278.)

So our mental images are sometimes wrong. Further, it is not obvious that we need mental images to answer visual or spatial questions. Computers can answer such questions quite accurately without drawing little pictures inside themselves. Can we demonstrate that mental images are at least sometimes useful, though, and that they have some of the properties we ordinarily associate with vision? Answering this question was one of the early triumphs of experimental research in cognitive psychology.

WHAT'S THE EVIDENCE?

Mental Imagery

Roger Shepard and Jacqueline Metzler (1971) conducted a classic study of how humans solve visual problems. They reasoned that, if people actually visualize mental images, then the time it takes them to rotate a mental image should be similar to the time it takes to rotate a real object.

Hypothesis When people have to rotate a mental image to answer a question, the farther they have to rotate it, the longer it will take them to answer the question.

Method The experimenters showed subjects pairs of two-dimensional drawings of three-dimensional objects, as in Figure 8.3 and asked whether the drawings in each pair represented the same object rotated in different directions or whether they represented different objects. (Try to answer this question yourself before reading further. Then, check answer B, page 278.)

The subjects could answer by pulling one lever to indicate *same* and another lever to indicate *different*. When the correct answer was *same*, a subject might determine that answer by rotating a mental image of the first picture until it matched the second. If so, the delay should depend on how far the image had to be rotated.

The delays before answering *different* were generally longer and less consistent than those for answering *same*—as is generally the case. Subjects could answer *same* as soon as they found a way to rotate the first object to match the second. However, to be sure that the objects were different, subjects might have to double-check several times and imagine more than one way of rotating the object.

Results Subjects were almost 97% accurate in determining both *same* and *different*. As predicted, their reaction time for responding *same* depended on the angular difference in orientation between the two views. For example, if the first image of a pair had to be rotated 30 degrees to match the second image, the subject needed a certain amount of time to pull the *same* lever. If the two images looked the same after the first one had been rotated 60 degrees, the subject took twice as long to pull the lever. In other words, the subjects reacted as if they were actually watching a little model of the object rotate in their head; the

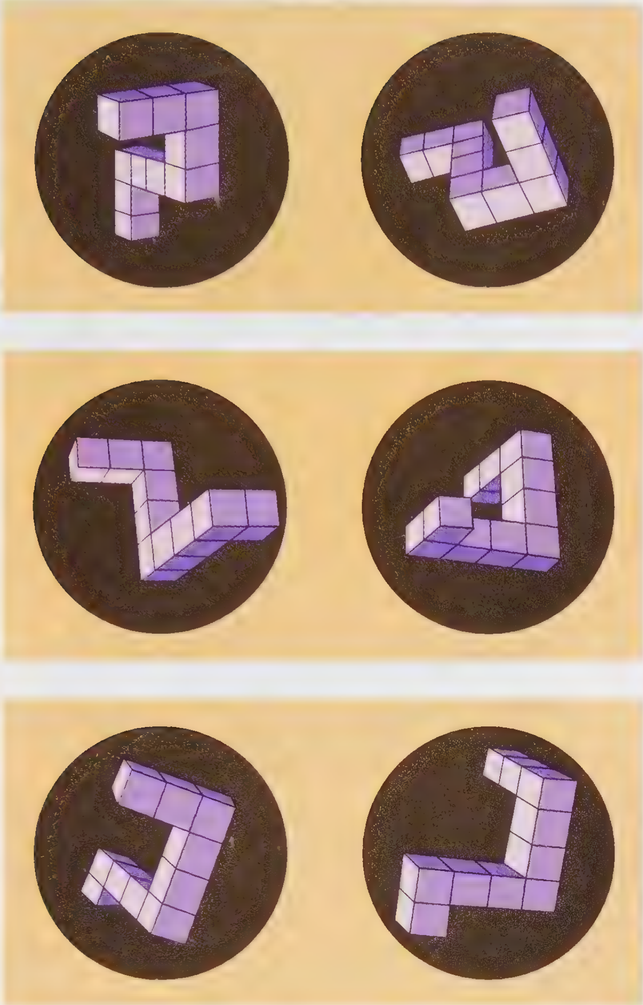


FIGURE 8.3 Examples of pairs of drawings used in an experiment by Shepard and Metzler (1971). Do the drawings for each pair represent the same object being rotated, or are they different objects? (See answer B on page 278.)

more the object needed to be rotated, the longer they took to determine the answer.

Interpretation Viewing a mental image is at least partly like real vision.

Whenever possible, researchers try to test a hypothesis in different ways. Even what appears to be an excellent experiment may have some hidden flaw or be subject to an alternative explanation. An alternative way of studying visual imagery is to ask people to imagine various objects while researchers record their eye movements. One study found that, while people were imagining objects, their eyes moved almost the same way as when they were actually looking at the same objects (Brandt & Stark, 1997). Even this study, of course, does not fully establish the relationship between imagined vision and real vision, but it helps to confirm a similarity.

* * *

SOMETHING TO THINK ABOUT

Some people report that they have auditory images as well as visual images. They "hear" words or songs "in their head." What kind of evidence would we need to test this claim? *

Using Mental Images: Cognitive Maps

You are staying at a hotel in an unfamiliar city. You walk a few blocks to get to a museum; then you turn and walk in another direction to get to a restaurant; after dinner, you turn again and walk to a theater. After the performance, how do you get back to the hotel? Do you retrace all of your steps? Can you find a shorter route? Or do you give up and hail a cab?

If you can find your way back, you do so by using a **cognitive map,** *a mental image of a spatial arrangement.* One way to measure the accuracy of people's cognitive maps is to test how well they can find the route from one place to another. Another way is to ask them to draw a map. As you might expect, people draw a more complete map of the areas they are most familiar with. When students try to draw a map of their college campus, they generally include the central buildings on campus and the buildings they enter most frequently (Saarinen, 1973). The longer students have been on campus, the more detail they include (Cohen & Cohen, 1985).

The errors people make in their cognitive maps follow some interesting patterns. First, they tend to remember street angles as being close to 90 degrees, even when they are not (Moar & Bower, 1983). We can easily understand that error. For practical purposes, all we need to remember is "go three blocks and turn left" or "go two blocks and turn right"; we do not burden our memory by recalling "turn 72 degrees to the right."

Second, people generally imagine geographic areas as being aligned neatly along a north-to-south axis and an east-to-west axis (Stevens & Coupe, 1978; B. Tversky, 1981). Try to answer these questions, for example: Which city is farther west—Reno, Nevada, or Los Angeles, California? And which is farther north—Philadelphia, Pennsylvania, or Rome, Italy? Most people reason that, because California is west of Nevada, Los Angeles is "obviously" west of Reno. (Figure 8.4 shows the true position of the cities.) Rome is in southern Europe, and Philadelphia is in the northern part of the United States; therefore, Philadelphia should be north of Rome, right? In fact, Rome is north of Philadelphia.

You now see the differences between a cognitive map and a real map: Cognitive maps, like other mental images, highlight some details, distort some, and omit others. Nevertheless, they are accurate enough for most practical purposes.

Attention

Much of intelligent behavior depends on successfully managing your attention. For example, while driving a car, you might devote most of your attention to the conversation, but

FIGURE 8.4 Logical versus actual: location of Reno and Los Angeles. Most people imagine that Los Angeles is farther west because California is west of Nevada.

if the traffic gets bad or if you have to look for a particular street, you might shift your attention more to your driving. You might even tell your friend to be quiet until you find your turn. People who either cannot shift their attention or shift it too easily, have trouble in school, on the job, and in everyday life.

Very complex tasks overload almost anyone's attention span. For example, Figure 8.5 shows a small part of the control room of the Three Mile Island nuclear power plant as it looked before the nearly disastrous accident in 1979. You notice immediately the enormous number of knobs and gauges; a nuclear power plant is, after all, a complicated system. What you cannot see in the picture is that, in certain cases, the knob that controlled a variable was in one place and the gauge measuring that variable was in another place. After the 1979 accident, ergonomists (human factors psychologists) helped redesign the plant so that mere humans could master the task of running it.

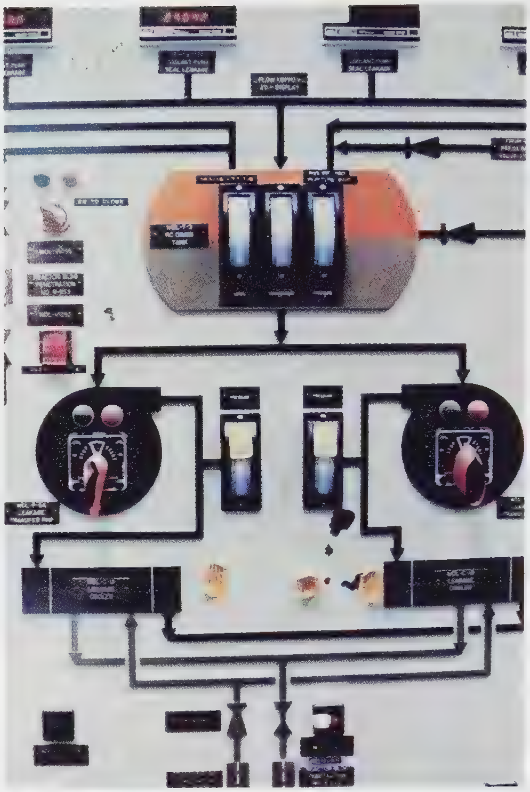


FIGURE 8.5 The Three Mile Island TMI-2 nuclear power plant had a complex and confusing control system, a small portion of which is shown here. Some of the important gauges were not easily visible to the operators, some of the gauges were poorly labeled, and many of the alarm signals had ambiguous meanings. The nuclear plant currently in operation at Three Mile Island has a much simpler and clearer operating system.

Preattentive and Attentive Processes

Sometimes you recognize familiar patterns automatically, without requiring any attention or working memory (Schneider & Shiffrin, 1977; Shiffrin & Schneider, 1977). For example, you might drive home along a familiar route, recognizing the landmarks and turning at the correct time even while thinking about something else. When you reach your driveway, you might realize that you do not even remember driving there. In other situations, you have to pay careful attention to what you are doing, one step at a time, as when you write an essay for your English class and have to tell your friends not to interrupt you until you finish.

TRY IT YOURSELF

To illustrate the difference between these two types of processing, first examine Figure 8.6. In each part of the figure, find the circle that is intersected by a vertical line.

You probably spotted the vertical line in *b* about as quickly as the vertical line in *a*, even though *b* has far more distracters (circles without lines) (Treisman & Souther, 1985). Apparently, people examine all the circles *in parallel.* That is, we can look at all the circles at once; finding the vertical line does not require attending to one circle at a time. Finding the line relies on a **preattentive process**—*a procedure for extracting information automatically and simultaneously across a large portion of the visual field* (Enns & Rensink, 1990).

Now look at Figure 8.7. Each part contains several pentagons, most of them pointing upward. Find the one pentagon in each part that points downward.

Most people take longer to find the pentagon pointing down in part *b* than in part *a*, because part *b* contains more distracters. The greater the number of distracters, the longer it takes to find the pentagon that is different; people must turn their attention to one pentagon at a time until they come to the correct one. In contrast to the preceding example, this task requires an **attentive process**—*a procedure that considers only one part of the visual field at a time.* An attentive process is ordinarily a *serial* process, because a person must attend to each part one after another in a series.

TRY IT YOURSELF

Here is another example of a preattentive or automatic process: Read the following instructions and then examine Figure 8.8 on page 276:

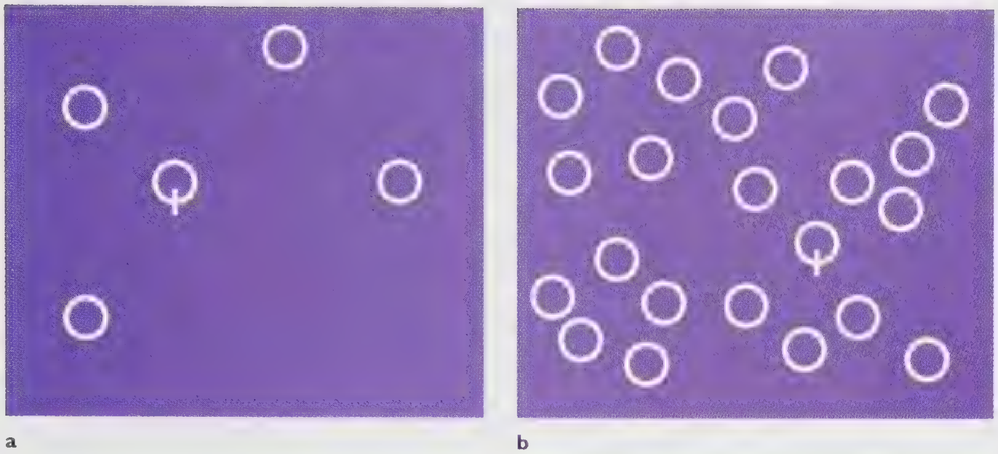


FIGURE 8.6 Demonstration of preattentive processes: Find the vertical line in each part. Most people find it about equally fast in both parts.

> Notice the blocks of color at the top of the figure. Scanning from left to right, give the name of each color as fast as you can. Then notice the nonsense syllables printed in different colors in the center of the figure. Don't try to pronounce them; just say the color of each one as fast as possible. Then turn to the real words at the bottom. Don't read them; quickly state the color in which each one is printed.

Most people find it very difficult not to read the words at the bottom of the figure. After all the practice you have had reading English, you can hardly bring yourself to look at the word RED, written in green letters, and say "green." *The tendency to read the word, instead of saying the color of ink as instructed,* is known as the **Stroop effect,** after the psychologist who discovered it. The effect is particularly strong when the discrepancy is small between the word and the color of the ink. That is, it is difficult enough to look at the word YELLOW in blue ink and say "blue"; it is even more difficult to look at YELLOW in orange ink and say "orange" (Klopfer, 1996).

With the Stroop effect, preattentive or automatic processing acts as interference, but in many situations we can harness it to our advantage. For example, if you were an ergonomist designing a piece of machinery, you could design warning signs that stand out by use of preattentive processes: The gauges in both the top and bottom rows of Figure 8.9 represent the current measurements of several items. If the machine is running properly, the first gauge should read about 70, the second should read about 40, the third should read about 30, and the fourth should read about 10. Note that, in the bottom row, all the safe readings point in the same direction, so an operator can detect a dangerous reading quickly, preattentively. In the top row, a person must attend to one gauge at a time, however.

The Attentional Blink

Sometimes you can do two things at once, especially if one of them is a simple, routine task such as driving your car down a familiar road. However, when any stimulus catches your attention, it briefly blocks your attention to anything else. Psychologists speak of a "perceptual bottleneck," implying that only one stimulus at a time can get through to your attentional processor (Pashler, 1994).

The clearest demonstration of this principle is a phenomenon called the **attentional blink,** which was named by analogy to a blink of the eyes; during the moment it takes you to blink, you cannot see anything. Similarly, ***during the moment after perceiving one stimulus, it is difficult to attend to something else.***

For example, consider the display shown in Figure 8.10. Participants watched a screen that briefly displayed two letters, one in green and one in red. A nonsense pat-

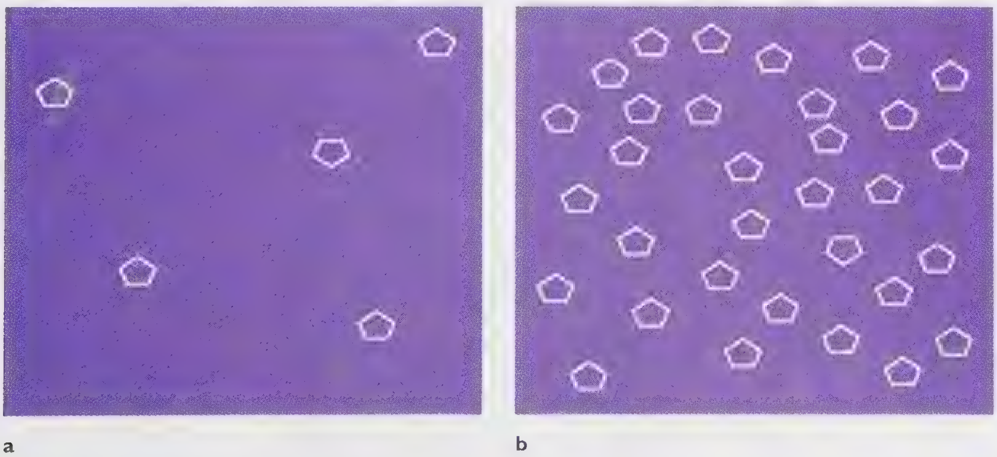


FIGURE 8.7 Demonstration of attentive processes: Find the pentagon pointing down in each part. Most people take longer to find it in part b.

FIGURE 8.8 Read (left to right) the color of the ink in each part. Try to ignore the words themselves. Your difficulties on the lowest part illustrate the Stroop effect.

tern followed to make the identification more difficult. If the two letters appeared on the screen simultaneously, participants could identify both of them with reasonable accuracy. However, if one letter preceded the other, participants named the first one correctly but had more trouble with the second one. The interference was especially severe with a delay of 200 milliseconds(ms) but still noticeable at a delay of 600ms (Duncan, Ward, & Shapiro, 1994). Similar results occur with many kinds of stimuli: Someone sees a letter or a word or hears a sound and then another one of the same kind; the result is decreased identification of the second one (Duncan, Martens, & Ward, 1997).

This period of several tenths of a second after presentation of the first stimulus is the attentional blink, the time when it is hard to register a second stimulus. What happens to information that is presented during the attentional blink? The brain does not shut it out completely. Evoked potentials recorded from the scalp indicate that the information reaches the cerebral cortex (Luck, Vogel, & Shapiro, 1996). Also, although you would ordinarily ignore a word or name you read during the attentional blink, you do notice it if it is your own name (Shapiro, Caldwell, & Sorensen, 1997).

CONCEPT CHECK

3. Suppose you are in a field full of brownish bushes and one brown rabbit that is not moving. If you want to find the rabbit, will you rely on attentive or nonattentive processes? What if the field has many stationary rabbits and one that is hopping? Will you then find it by attentive or nonattentive processes? (Check your answer on page 278.)

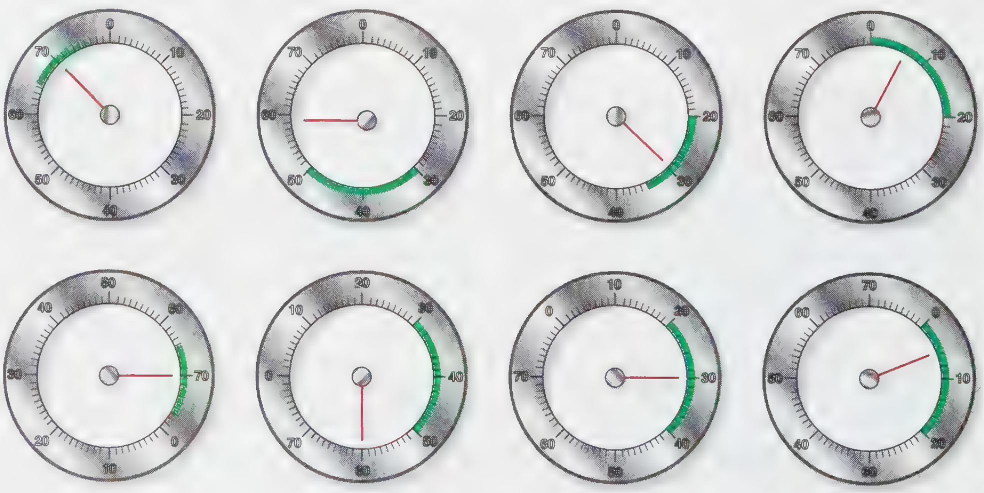


FIGURE 8.9 Each gauge represents a measurement of a different variable in a machine, such as an airplane. The top row shows one way of presenting the information. The operator must check each gauge one at a time to find out whether the reading is within the safe range for that variable. The bottom row shows the information represented in a way that is easier to read. The safe range for each variable is rotated to the same visual position. At a glance, the operator can detect any reading outside the safe zone.

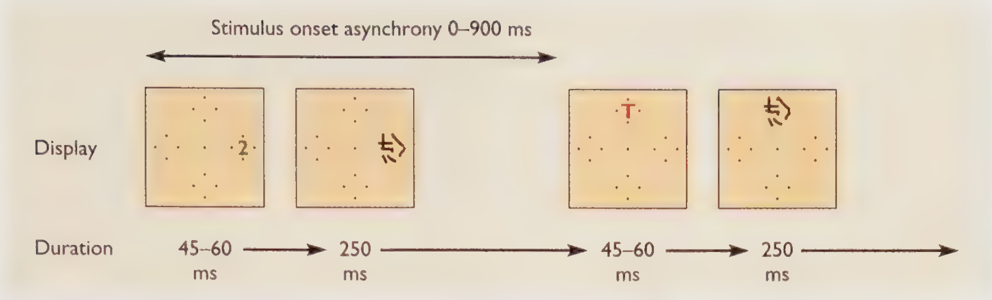


FIGURE 8.10 Participants watched a screen that showed a green number (2 or 5) and a red letter (L or T). Sometimes they appeared simultaneously, sometimes with a delay of up to 900 milliseconds between them. Immediately after each stimulus disappeared, an interfering pattern took its place. The results demonstrated interference from the first stimulus upon the second stimulus. (Modified from Duncan, Ward, & Shapiro, 1994).

THE MESSAGE *Thinking as a Laboratory Phenomenon*

Throughout this module, we have considered experiments from which psychologists drew inferences about people's categorization, conceptual networks, cognitive maps, and attention. Drawing inferences is a time-honored way of doing science: Physicists describe the properties of electrons and quarks; astronomers estimate the age of the universe; biologists reconstruct the probable structure of dinosaurs and other species long since extinct. When dealing with unobservables—whether it be attention processes or quarks—we should remember that any one line of evidence is tentative. The conclusions never become certain, but they become stronger if we find many kinds of studies, using different methods, whose results all point in the same direction.

SUMMARY

✷ *Categorization.* People use many categories for which we have no clear or simple definition; we determine whether something fits the category by how closely it resembles a few familiar examples. (page 269)

✷ *Conceptual networks.* We store representations of words or concepts with links to related concepts. Hearing or thinking about one concept will temporarily prime the linked concepts. (page 270)

✷ *Mental imagery.* Mental images resemble vision in certain respects. For example, the time required to answer questions about a rotating object depends on how far the object would actually rotate and, therefore, the necessary time. (page 272)

✷ *Cognitive maps.* People learn to find their way by using cognitive maps, but they make certain consistent errors in these maps, such as remembering all turns as being close to 90-degree angles. (page 273)

✷ *Attentive and preattentive processes.* We notice some items—such as a straight line among circles or a moving object among stationary ones—almost at once, in spite of many potential distracters. Noticing other, less distinct items requires more careful attention to one possible target after another. However, this difference is one of degree; all tasks require attention, although some require more than others. (page 274)

✷ *Automatic attention.* Sometimes it is difficult to avoid attending to certain stimuli; for example, it is difficult to state the color of the ink in which words are written while ignoring the words themselves (especially if they are color names). (page 275)

✷ *Attentional blink.* Immediately after perceiving a stimulus, there is a brief moment when it is difficult to perceive other stimuli, although they may exert subtle effects on later perceptions. (page 275)

Suggestion for Further Reading

Johnson-Laird, P. N. (1988). *The computer and the mind.* Cambridge, MA: Harvard University Press. Description of cognitive psychology and its relationship to computer science, written in nontechnical language.

Terms

cognition the processes of thinking, gaining knowledge, and dealing with knowledge (page 269)

prototype a familiar or typical example of a category (page 270)

spreading activation the process by which the activation of one concept also activates or primes other concepts that are linked to it (page 271)

cognitive map a mental representation of a spatial arrangement (page 273)

preattentive process a procedure for extracting information automatically and simultaneously across a large portion of the visual field (page 274)

attentive process a procedure that extracts information from one part of the visual field at a time (page 274)

Stroop effect the tendency to read a word, especially if it is a color name, in spite of instructions to disregard the word and state the color of the ink in which it is printed (page 275)

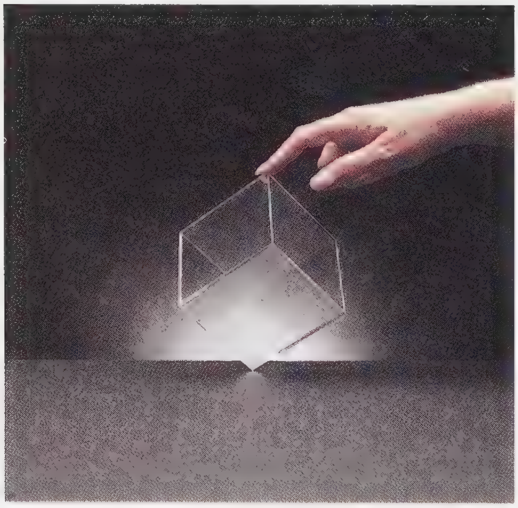
attentional blink a brief period after perceiving a stimulus, during which it is difficult to attend to another stimulus (page 275)

Answers to Concept Checks

1. It would take longer to answer whether fashion models sometimes get sick. Wearing dresses is a distinctive feature of fashion models; becoming ill is not. To answer the second question, you have to reason that models are people, and people sometimes get sick. (page 271)
2. People should answer *alarm clock* or *cuckoo clock* faster to the second question. The cue "tells time" strongly activates only a few concepts and therefore provides a strong start, whereas the cue "makes sounds" activates so many concepts that it only very weakly primes the correct answer. (page 272)
3. Finding a motionless brown rabbit in a field full of brown objects will require attentive processes, but you could use preattentive processes to find a hopping rabbit in a field where nothing else is moving. (For this reason, small animals that are in danger of predation stay motionless when they can.) (page 276)

Answers to Other Questions in the Text

A. The cube has six (*not* four) remaining corners. (page 272)



B. The objects in pair (a) are the same; in (b), they are the same; and in (c), they are different. (page 273)

Web Resources

The Stroop Effect
www.conex.com.br/user/hermann/eng/stroop.htm
This student-created site has a nice Java applet for demonstrating the Stroop effect. Test yourself, or a friend or classmate, for accuracy and time: Does either improve with practice?

Problem Solving, Expertise, and Error

What do experts know or do that sets them apart from other people?

How can we improve our ability to solve problems?

Why do people sometimes reason illogically?

On a college physics exam, a student was once asked how to use a barometer to determine the height of a building. He answered that he would tie a long string to the barometer, go to the top of the building, and carefully lower the barometer until it reached the ground. Then, he would cut the string and measure its length.

How would you carry 98 water bottles—all at one time, with no wheelbarrow or truck? When faced with a new problem, sometimes people find a novel and effective solution, and sometimes they do not.

When the professor marked this answer incorrect, the student asked why. "Well," said the professor, "your method would work, but it's not the method I wanted you to use." When the student objected, the professor offered, as a compromise, to let the student try again.

"All right," the student said. "Take the barometer to the top of the building, drop it, and measure the time it takes to hit the ground. Then, from the formula for the speed of a falling object, using the gravitational constant, calculate the height of the building."

"Hmmm," replied the professor. "That too would work. And it does make use of physical principles. But it still isn't the answer I had in mind. Can you think of another way?"

"Another way? Sure," replied the student. "Place the barometer next to the building on a sunny day. Measure the height of the barometer and the length of its shadow. Also measure the length of the building's shadow. Then, use the formula

$$\frac{\text{height of barometer}}{\text{length of barometer's shadow}} = \frac{\text{height of building}}{\text{length of building's shadow}}$$

The professor was becoming more and more impressed, but he was still reluctant to give credit for the answer. He asked for yet another method.

The student suggested, "Measure the barometer's height. Then walk up the stairs of the building, marking it off in units of the barometer's height. At the top, take the number of barometer units and multiply by the height of the barometer to get the height of the building."

The professor sighed: "Just give me one more way—any other way—and I'll give you credit, even if it's not the answer I wanted."

"Really?" asked the student with a smile. "Any other way?"

"Yes, any other way."

"All right," said the student. "Go to the man who owns the building and say, 'Hey, buddy, if you tell me how tall this building is, I'll give you this neat barometer!' "

We sometimes face a logical or practical problem that we have never tried to solve before. We must devise a new solution; we cannot rely on a memorized or practiced solution. Sometimes people develop creative, imaginative solutions, like the ones that the physics student proposed. Sometimes they offer less imaginative, but still reasonable, solutions. Other times they suggest something quite illogical or offer no solution at all. Psychologists study problem-solving behavior partly to understand the thought processes behind it and partly to look for ways to help people reason more effectively.

Expertise

People vary in their performance on problem-solving and decision-making tasks. In the barometer story just described, we would probably talk about the student's creativity; in other cases, we might talk of someone's expertise. In either case, some people seem more able than others to understand a problem and to find feasible solutions.

Expert performance in music, art, athletics, and other fields can be extremely impressive. An expert crossword puzzle solver not only completes the *New York Times* Sunday crossword—an amazing feat in itself—but finds it too easy and has to have a race with another expert puzzle solver to make the task interesting. An expert birdwatcher can look at a small photo that is slightly out of focus and identify not only the species of bird, but also the subspecies, the sex, the age, and whether the bird is in summer or winter plumage. An expert at the game of bridge will say something like, "It was obvious from the bidding that East had to have the king of diamonds, so obviously it was better to try for an endplay than a finesse." (And a casual bridge player like me wonders "Huh? How was any of that obvious?")

Practice Makes Perfect (or Nearly Perfect)

Confronted by such amazing skills, it is tempting to assume that these experts were born with a special talent. Not so, say psychologists who have studied expertise (Ericsson & Charness, 1994; Ericsson, Krampe, & Tesch-Römer, 1993). In virtually every kind of endeavor, people need years of concentrated practice before they emerge as experts. For most kinds of performance, ranging from chess to sports to violin playing, the general rule is that one needs about ten years of concentrated practice. For decades, the Russians have dominated world chess competitions. Is that because Russians have a special talent for chess? Probably not; the more parsimonious explanation is that more Russians play chess, beginning in childhood, than people in other countries. In general, researchers find, if you know how many people in a given country play chess, you can predict fairly accurately how good the best player is in that country (Charness & Gerchak, 1996).

Hungarian author Laszlo Polgar set out to demonstrate his conviction that almost anyone can become an expert at something, with sufficient effort. He allowed his three young daughters to explore several fields; when they showed an interest in chess, he devoted enormous efforts to nurturing their chess skills. Today, all three daughters are outstanding chess players. One of them, Judit Polgar, reached grand master status about ten years after she started studying the game. (See Figure 8.11.)

Similarly, most of the world's best violinists began learning the violin in early childhood and continue practicing 3–4 hours a day throughout their lives. The top tennis and golf players also spend many hours practicing. Those hours of practice do not include performances; while performing, one does not have the opportunity to correct one's mistakes. Thus, an expert violinist practices the especially difficult passages; a tennis player spends hours working on the backhand shot; a golfer spends hours practicing chip shots. Practice is even more effective if a coach is present to provide extra feedback.

FIGURE 8.11 Judit Polgar confirmed her father's confidence that prolonged effort could make her an expert in her chosen field, chess. By reaching the status of grand master at age 15 years and 5 months, she beat Bobby Fisher's previous record for being the youngest.

Some researchers have argued that becoming an expert depends *entirely* on practice and is independent of any inborn talents or predispositions. That claim is almost certainly an overstatement (Gardner, 1995). If ten people began developing a skill at exactly the same age and practiced for exactly the same amount of time for the same number of years, would they all reach exactly the same level of expertise? Probably not. At a minimum, we must admit the obvious: A person with genes for being short and slow will not become an expert basketball player, a person born blind will not become an expert photographer, and so forth. Furthermore, in any field, those who show early success are most likely to devote the necessary practice time to become experts. The main point, however, is that, in any field, most of the difference between the experts and the merely average-to-good performers depends on effort and practice.

So, does all this mean that *you* could become an expert at something? Well, if you have waited until you are college age to start on a skill as competitive as chess, violin, or basketball, you are very late; nearly all experts in those fields were early starters. Still, if you choose an appropriate field and devote enough effort, you can become an expert at something. However, do not underestimate the effort and sacrifice that will be necessary. For Judit Polgar to become a grand master at chess, she devoted about 8 hours a day to chess, from age 5 to 15. She did not go to public school, so

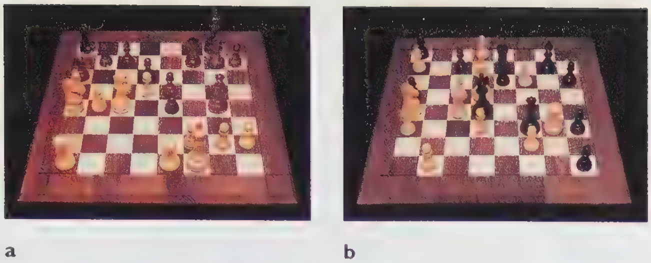


FIGURE 8.12 Pieces arranged on a chessboard as they might actually occur in a game (a) and in a random manner (b). Master chess players can memorize the realistic pattern much better than average players can, but they are no better than average at memorizing the random pattern.

she missed most of the usual childhood activities. Whatever your chosen field, if you want to excel in it, you will have to commit yourself to an enormous amount of work.

Expert Pattern Recognition

Once someone has become an expert, what exactly does that person do that other people do not? In other words, what makes someone an expert? One important characteristic of experts is that they can look at a pattern and recognize its important features quickly. Much of the research in this area deals with chess, because it is easier to identify the experts in chess than in most other fields. In a typical experiment (de Groot, 1966), people were shown pieces on a chessboard, as in Figure 8.12, for 5 seconds. Then, they were asked to recall the position of all the pieces. When the pieces were arranged as they might occur in an actual game, expert players could recall 91% of the positions, whereas novices could recall only 41%. When the pieces were arranged randomly, however, the expert players did no better than the nonexperts. That is, on the average, expert chess players are not superior in overall memory or intelligence; they have simply learned to recognize and remember a great many of the most common chessboard patterns.

Further evidence for this conclusion includes the fact that the top-level grand master chess players can play simultaneous games against six or so other highly ranked opponents and play almost as well as if they were facing only one opponent (Gobet & Simon, 1996). Simultaneous play offers little opportunity for planning five or six moves ahead and for considering all the possible countermoves by one's opponent; in simultaneous play, one is forced to rely on quickly recognizing a pattern and knowing what is a good move for that situation.

Problem Solving

Can people learn general skills of problem solving, which they could apply to new and unfamiliar questions? To some extent, yes, they can (Bransford & Stein, 1984).

Generally, we go through four phases when we set about solving a problem (Polya, 1957): (1) understanding the problem, (2) generating one or more hypotheses, (3) testing the hypotheses, and (4) checking the result (Figure 8.13). A scientist goes through those four phases when approaching a new, complex phenomenon, and you would probably go through these phases when trying to assemble a bicycle that came with garbled instructions. We shall discuss these four phases of problem solving in detail.

Understanding and Simplifying a Difficult Problem

You are facing a question or a problem, and you have no idea how to begin. You may even think the problem is unsolvable. Then someone shows you how to solve it, and you realize, "I could have done that, if I had only thought of trying it that way."

When you do not know how to solve a problem, try starting with a simpler version of it. For example, here is what may appear to be a difficult, even impossible, problem:

> A professor hands back students' test papers at random. On the average, how many students will accidentally receive their own paper?

Note that the problem does not specify how many students are in the class. At first, you may not know how to approach the problem, but see what happens if you start with some simpler cases: How many students will get their own paper back if there is only one student in the class? One, of course. What if there are two students? There is a 50% chance that both will get their own paper back and a 50% chance that neither will. On the average, one student will get the correct paper. What if there are three students? Each student then has one chance in three of getting his or her own paper. A one-third chance times three students means that, on the average, one student will get the correct paper. Already you can see the pattern: If there are 100 students,

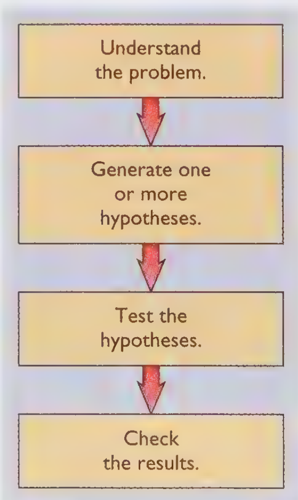


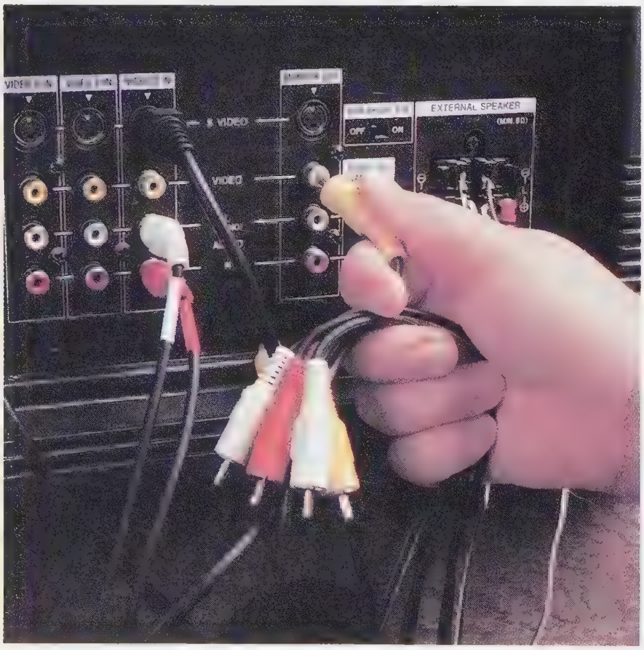
FIGURE 8.13 The four steps to solving a problem.

each student has one chance in a hundred of getting his or her own paper back. No matter how many students are in the class, on average, one student will get his or her own paper back.

Generating Hypotheses

After you have simplified a problem as much as possible, you need to generate *hypotheses*—preliminary interpretations that you can evaluate or test. In some cases, you can generate a complete list of possibilities and then test them all. For example, suppose you want to connect your television set to a pair of stereo amplifiers and a VCR or DVD player, but you have lost the instruction manuals (or you refuse to read them). You have several cables to attach, and each device has some input and output channels. You could simply connect the cables by trial and error, testing every possibility until you find one that works. *A mechanical, repetitive mathematical procedure for solving a problem* is called an **algorithm.** Testing every possible hypothesis is one example of an algorithm. The rules for alphabetizing a list are another example. You can also learn an algorithm for how to win, or at least tie, every time you play tic-tac-toe.

In many situations, however, you could not list all the hypotheses, much less test them. For example, consider the question, "What should I do with my life?" Confronted with a vast and confusing array of possibilities, you must simplify the task. **Heuristics** are *strategies for simplifying a problem or for guiding an investigation.* In this career-choice example, you might limit yourself to a few possible careers that tap your interests and skills and then try to learn as much as you can about those few careers.



To find the right way to connect the lines, you could use the simple algorithm of trying every possible combination, one after the other.

To illustrate the contrast between algorithms and heuristics, consider chess. At a typical point in a game of chess, a player has about 25 legal moves, for each of which the opponent has 25 legal replies. If you wanted to choose the best possible move by an algorithm, you would consider each of your 25 possible moves, each of your opponent's possible replies, each of your possible next moves, and so forth—say, five or six moves ahead. Finally, you would select the move that gives you the best result, assuming that your opponent made the best possible reply at each point. This algorithm, however, will overburden your memory, so you simplify it with some heuristics: On each move, you select just a few possible moves for serious consideration, aided by your recognition of similar positions you have faced in past games. You consider just a few of your opponent's likely responses, a few of your possible next moves, and so forth. If you decide that one of your possible moves would be disastrous, you do not waste time reconsidering it after your opponent's move.

In contrast, the best chess-playing computers do rely on algorithms to test every possible move. The Deep Blue computer program that beat Garry Kasparov, the best human player, did so by considering every possible move and every possible reply, 14 or 15 moves ahead, and as many as 45 moves ahead when necessary (Mechner, 1998). The computer succeeded because it can consider 200 million positions per second (and remember the outcomes). With such a powerful algorithm, no heuristics are necessary.

However, in the game of Go (Figure 8.14), which has a 19 × 19 grid and very complex strategies, each player has an average of about 250 possible moves, for each of which the opponent would have 250 possible replies. Whereas

FIGURE 8.14 In the Chinese game Go, opponents alternate placing their markers on a 19 × 19 grid, attempting to arrange their own pieces in a safe arrangement or to surround opposing pieces that are not in safe arrangements. For this game, an algorithm of imagining every possible move and then every possible reply would be extemely inefficient; effective strategies require heuristics to restrict the moves under consideration.

Deep Blue can run its algorithm to find the best possible chess move in 3 minutes, a computer would need an estimated 70 years to complete a comparable algorithm for Go! At the time that Deep Blue beat Kasparov, the best available computer program for playing Go was at the level of a beginner (Mechner, 1998). Even if we assume that computers will continue to get faster and more powerful, programming a computer to play Go successfully will probably require developing some heuristics—in effect, programming the computer to think more like a human.

CONCEPT CHECK

4. Suppose you are a traveling salesperson. You must start from your home city, visit each of several other cities, and then return home. Your task is to find the shortest route. What would be the appropriate algorithm? What would be a possible heuristic for simplifying the problem? (Check your answer on page 295.)

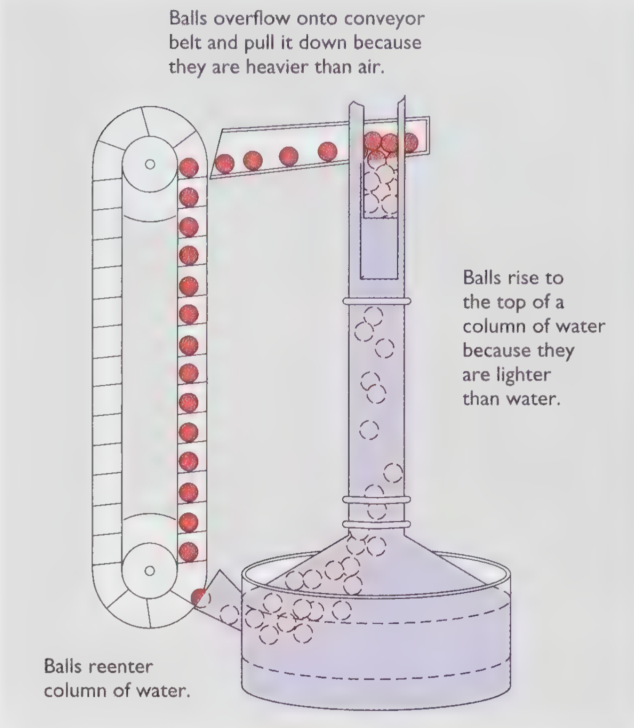


FIGURE 8.15 What is wrong with this perpetual motion machine?

Testing Hypotheses and Checking the Results

If you think you have solved a problem, test your idea to see whether it will work. Many people who think they have a great idea never bother to try it out, even on a small scale. One inventor applied for a patent on the "perpetual motion machine" shown in Figure 8.15. Rubber balls, being lighter than water, rise in a column of water and flow over the top. The balls are heavier than air, so they then fall, thus moving a belt and thereby generating energy. At the bottom, they reenter the water column. Do you see why this system could never work? You would if you tried to build it. (Check your answer on page 296, answer C.)

The final step for solving a problem is to check and recheck the results, or at least to find out whether the result is plausible. For example, one article in the journal *Science* reported that fields in California's Imperial Valley produce 750,000 melons per acre. One reader wrote to the journal to point out that 750,000 melons per acre meant about 17 melons per square foot. With tongue in cheek, he asked whether the weight of all those melons might be causing California's earthquakes (Hoffman, 1992).

Generalizing Solutions to Similar Problems

After laboriously solving one problem, can people then solve a related problem more easily? Can they at least recognize that the new problem is related to the old problem, so that they know where to start?

Sometimes, but all too frequently they do not. Many people who understand the laws of probability fail to see how those laws might apply to real-life situations (Nisbett, Fong, Lehman, & Cheng, 1987). For example, most people who flipped a coin 10 times and got 10 consecutive heads would not expect more than 5 heads out of the next 10 flips. But the same people might expect a basketball team that won 10 consecutive games to win the next 10 games as well. The basketball situation is not exactly the same as coin flipping, but it does have some similarities: A long winning streak depends partly on chance.

In other situations as well, people who have solved one problem correctly fail to solve a second problem that is basically similar, unless someone explains that the two problems are similar (Gick & Holyoak, 1980). For example, Figure 8.16a shows a coiled garden hose. When the water spurts out, what path will it take? (Draw it.) Figure 8.16b shows a curved gun barrel. When the bullet comes out, what path will it take? (Draw it.)

Almost everyone draws the water coming out of the garden hose in a straight path. Even after doing so, however, many people draw a bullet coming out of a gun in a curved path, as if the bullet remembered the curved path it had just taken (Kaiser, Jonides, & Alexander, 1986). The physics is the same in both situations: Except for the effects of gravity, both the water and the bullet will follow a straight path.

Sometimes we recognize similar problems and use our solution to an old problem as a guide to solving a new one (Figure 8.17), but sometimes we do not. What accounts for the difference? It is easier to generalize a solution after we have seen several examples of it; if we have seen only a single example, we may think of the solution in only that one context (Gick & Holyoak, 1983).

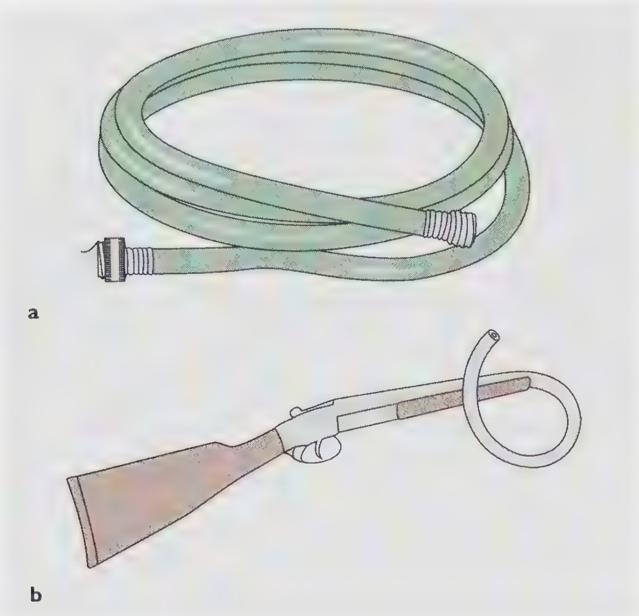


FIGURE 8.16 (a) Draw the trajectory of water as it flows out of a coiled garden hose. (b) Draw the trajectory of a bullet as it leaves a coiled gun barrel.

FIGURE 8.17 The computer mouse was invented by a computer scientist who was familiar with an engineering device called a planimeter that he believed could be modified for use with computers. Such insights are unusual; most people do not generalize a solution from one task to another.

Special Features of Insight Problems

Some of the problems we have been discussing are "insight" problems or "aha!" problems—the kind in which the correct answer occurs to you suddenly or not at all. Here is a clear example of an insight problem (Gardner, 1978): Figure 8.18 shows an object that was made by just cutting and bending an ordinary piece of cardboard. How was it made? If you think you know, take a piece of paper and try to make it yourself. (The correct answer is on page 296, answer D.)

TRY IT YOURSELF

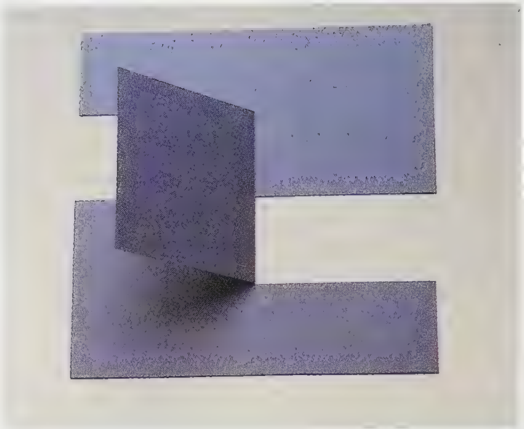


FIGURE 8.18 This object was made by cutting and folding an ordinary piece of cardboard, with nothing left over. How was it done?

Insightful Problem Solving: Sudden or Gradual?

Solving insight problems differs from solving, say, algebra problems. Most people can look at an algebra problem and rather accurately predict whether they will be able to solve it, and if so how quickly. As they work on the problem, they can estimate how close they are to reaching a solution. On insight problems, however, people give inaccurate estimates about whether they are about to solve the problem or not (Metcalfe & Wiebe, 1987). Frequently, someone will say, "I have no idea whether I will ever solve this problem," and then suddenly announce the correct answer a minute or so later.

So, it appears that the answer comes suddenly, or not at all. But does it, really? If you were groping your way around in a dark room, you would have no idea how soon you were going to find the door, but that does not mean that you had made no progress. You would have learned much about the room, including many places where the door was *not.*

So, maybe people are making progress without realizing it, when they struggle with insight problems. To test this possibility, psychologists gave students problems having the following form:

The following three words are all associated with one other word. What is that word?

color numbers oil

In this case, the correct answer is *paint.* As with other insight questions, subjects reported that the answer came to them suddenly or not at all and that they could not tell whether they were about to think of the answer or not. Then, the experimenters gave the subjects paired sets of three words each, like those shown here in Sets 1 and 2, to examine for 12 seconds. In each pair, one set had a correct answer (like *paint* in the example just given). For the other set, no one word was associated with all three items. Subjects were asked to generate a

TRY IT YOURSELF

correct answer if they could; if not, they were to guess *which* set had a correct answer and to say how confident they were of their guess. Examples:

Set 1

playing	credit	report	*or*	still	pages	music

Set 2

town	root	car	*or*	ticket	shop	broker

(You can check your answers on page 296, answer E.)

The main result was that, when subjects could not find the correct answer, they still guessed with greater than 50% accuracy which set had a correct answer (Bowers, Regehr, Balthazard, & Parker, 1990). Even when subjects said they had "no confidence" at all in their guesses, they were still right more often than not. In short, insight solutions are not as sudden as they seem; people may be approaching a solution without even realizing it.

The Characteristics of Creative Problem Solving

Solving a problem or developing a new product is always a creative activity, but some solutions and products seem more creative than others. Psychologists frequently define **creativity** as ***the development of novel, socially valued products*** (Mumford & Gustafson, 1988). In principle this is a reasonable definition, although in practice it can be hard to apply; many unusual works of art, music, and literature were very lightly regarded in their own time but later hailed as classics. By the usual definition of *creativity,* these works became creative many decades after they were produced!

Could you or I develop a great scientific theory, if we addressed the right question and had all the necessary information? If you want to try this challenge, examine the data in the table below. Five pairs of numbers, arbitrarily labeled *s* and *q,* are measurements of two physical variables. I won't tell you what the variables are, because you might be familiar with the theory in question. Examine each of the pairs of variables to see whether you can find a single mathematical equation relating *s* to *q*.

TRY IT YOURSELF

In one study, 4 of 14 university students managed to find the correct equation within one hour, thus reproducing a great scientific discovery (Qin & Simon, 1990). You may use a calculator. Recommendation: Either copy these data onto another sheet of paper or cover the text that follows so that you do not accidentally read the correct formula.

s	*q*
36	88
67.25	224.7
93	365.3
141	687
483.8	4332.1

The data represent measurements regarding the first five planets of our solar system. Column *s* gives the distance from the sun in millions of miles; column *q* gives the time (in Earth days) required for rotation around the sun. The German astronomer Johannes Kepler (1571–1630) is regarded as a genius and as the founder of modern astronomy for discerning the relationship between these two sets of data.

The correct formula can be expressed in any of the following ways:

$$s^3/q^2 = 6.025$$
$$s^3/6.025 = q^2$$
$$q^{2/3} = 0.55\,s$$
$$s^{1.5} = 2.45\,q$$

That is, the cube of the distance from the sun is related to the square of the period of rotation. The fact that 4 university students solved this problem, in far less time than Kepler needed, indicates that the actual problem solving requires only good mathematical skill and persistence, not a special talent available to only one person in a million. Kepler was a true genius because he guessed that there *was* a relationship between the two sets of numbers: He guessed that it was a solvable problem.

Creative problem solving, evident in this temporary bridge made of old railroad cars, has two elements: novelty and social value.

SOMETHING TO THINK ABOUT

Are people of normal talents also capable of great creative achievements in art and literature? How could we test that ability? ✷

The Characteristics of Creativity

Are some people naturally more creative than other people are? The Torrance Tests of Creative Thinking measure creativity using items similar to the one shown in Figure 8.19 (Torrance, 1980, 1981, 1982). We know that such tests measure something consistent; if you took tests of this type on several occasions, you would probably get approximately the same score each time. Several other tests have also been devised for measuring creativity of various sorts. The best summary is that there are many kinds of creativity: You might be creative in art but not in writing, or creative in writing but not in scientific problem solving (Sternberg & Lubart, 1996).